

YUJIN HAM

Rice University, 6100 Main Street MS 366, Houston, TX 77005
Yujin.Ham@rice.edu | Personal Website [↗](#)

RESEARCH INTERESTS

My research interests are in computer vision and 3D perception, with a focus on large-scale 3D/4D scene reconstruction, scene understanding, and generative vision for dynamic real-world environments, using vision foundation models.

EDUCATION

Rice University Ph.D. in Electrical and Computer Engineering	Houston, TX, United States Aug 2022 – Dec 2027 (Expected)
• Advisor: <i>Prof. Guha Balakrishnan</i> ↗	
Ewha Womans University M.S. in Electronic and Electrical Engineering	Seoul, South Korea Mar 2020 – Feb 2022
• Advisor: <i>Prof. Je-Won Kang</i> ↗	
• Thesis: Quality-adaptive Image Compression Artifact Removal using Deep Learning	
Ewha Womans University B.S. in Electronics and Electrical Engineering	Seoul, South Korea Mar 2016 – Feb 2020

RESEARCH EXPERIENCE

Rice Vision and Imaging Group (RVIG) Advisor: <i>Prof. Guha Balakrishnan</i> ↗ Research Assistant, Rice University	Aug 2022 – Present
• Generative vision models: Developed a generative video editing framework for removing people from egocentric walking-tour videos and synthesizing human-free environment walkthroughs by fine-tuning a vision foundation model on a curated data pipeline; first author, CVPR 2026	
• Multi-camera 3D vision: Developed DRAGON, a Gaussian Splatting-based registration framework integrating aerial and ground imagery for 3D building reconstruction; first author, ICCP 2024 (Oral)	
• Large-scale 3D reconstruction (WRIVA): Enhanced multi-altitude 3D reconstruction using diffusion-based refinement and ViT-based perceptual regularization in a multi-team IARPA project	
Samsung Research America (SRA) Mentor: <i>Abhinav Kumar, Ph.D.</i> ↗ Research Intern, Mobile Processor Innovation (MPI)	May 2026 – Aug 2026
• Developed a method for AI-based computational imaging for video enhancement using generative models; work resulted in a patent application and a paper in preparation	
Information Coding and Processing Lab. (ICPL) Advisor: <i>Prof. Je-Won Kang</i> ↗ Research Assistant, Ewha Womans University	Mar 2020 – Feb 2022
• Developed deep learning methods for compression artifact reduction and image enhancement, including domain adaptation approaches	

PROJECTS

Walk-through Rendering from Images of Varying Altitude (WRIVA) <i>Intelligence Advanced Research Projects Activity (IARPA)</i> ↗	Dec 2022 – Present
• Enhanced novel-view rendering quality across varying altitudes using diffusion-based refinement conditioned on scene-matched reference images	
• Added a DreamSim (ViT-based) perceptual regularization loss that penalizes perceptual distance between neighboring-viewpoint renders, improving 3D reconstruction through multi-view consistency	
• Contributed to the team's Airflow DAG pipeline for independent program evaluation; participated in code-freeze, internal evaluation, and patching cycles	
• Packaged my pipeline components into a shared Docker image used across sub-teams	

Compression and Transmission Technologies for Ultra High Quality Immersive Videos Supporting 6DoF

Mar 2020 – Dec 2020

Institute of Information & Communications Technology Planning & Evaluation (IITP) [↗](#)

- Developed deep learning methods for quantization-noise removal in 6DoF immersive video

PUBLICATIONS

1. **Y. Ham**, J. Kim, V. Boominathan, G. Balakrishnan, “**Generating Humanless Environment Walkthroughs from Egocentric Walking Tour Videos**”, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [↗](#)
2. J. Murrugarra-Llerena, P. Chitale, Z. Liu, K. Ao, **Y. Ham**, G. Balakrishnan, P. Cascante-Bonilla, “**EgoGroups: A Large-Scale Dataset for Detecting Groups of People in the Wild**”, *arXiv preprint arXiv:2603.22249*, 2026.
3. **Y. Ham**, M. Michalkiewicz, G. Balakrishnan, “**DRAGON: Drone and Ground Gaussian Splatting for 3D Building Reconstruction**”, *International Conference on Computational Photography (ICCP)*, 2024. [↗](#)
4. **Y. Ham**, C. Yoo, J. Kang, “**Training Compression Artifacts Reduction Network with Domain Adaptation**”, *Applications of Digital Image Processing XLIV (SPIE)*, 2021. [↗](#)
5. N. Kim, **Y. Ham**, J. Kang, “**Effective Video Captioning Algorithm Using Feature Attention Model**”, *Image Processing and Image Understanding*, 2020.

SKILLS

Programming/Tools: Python, C++, MATLAB, Git, Docker, Airflow
ML/DL: PyTorch, TensorFlow, Hugging Face, Diffusers
Computer Vision/3D: OpenCV, COLMAP, Nerfstudio, 3D Gaussian Splatting
Languages: English (fluent), Korean (native)

HONORS AND AWARDS

Ignite Entrepreneurship Trek to Silicon Valley ↗ Rice University	2025
SWITCH Mentoring Scholarship ↗ Rice University	2024
ICCP 2024 Student Travel Award US National Science Foundation (NSF) & Swiss NSF	2024
Rice University Department of Electrical and Computer Engineering Fellowship Rice University	2022
Grace Hopper Celebration (GHC) Scholarship 2021 AnitaB.org ↗ & Association for Computing Machinery	2021
Admission Scholarship for Outstanding Scientists Ewha Womans University	2020
Student Portfolio Award (1st prize) Accreditation Board for Engineering Education of Korea (ABEEK)	2019
Global Frontier Travel Grant (Visiting the US for 2 weeks) Ewha Womans University	2018

TEACHING EXPERIENCE

Teaching Assistant Rice University <i>ELEC 542: Generative AI for Image Data</i>	Fall 2023, Fall 2026
SWITCH Graduate Mentor Rice University <i>Summer Web-Based Institute for Technologies in CompSci and Healthcare (SWITCH)</i> ↗	Summer 2024
Teaching Assistant / Peer Instructor Ewha Womans University <i>Random Process (35477-01), Digital Engineering (30266-01)</i>	2018 – 2021